

Statement of Work
Project Name: Aven Tracker

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Revision History

Revision	Change History
1.0	Initial Draft

Abbreviations and Acronyms

SOW	–	Statement of Work
MS	–	Milestone
MCU	–	Micro controller Unit
RFQ	–	Request for Quotation
PCB	–	Printed Circuit Boards
CAD	–	Computer Aided Design
ATP	–	Acceptance Test Procedure
PDR	–	Preliminary Design Review
TBD	–	To Be Decided
GUI	–	Graphical User Interface
PCB	–	Printed Circuit Board

1. Scope of Document

The purpose of this Statement of Work (SOW) document is to define the scope of work, schedule for design, development and deliverables of **Aven Tracker**.

Trinova AI is going to design and develop a prototype of **Aven Tracker**. The device consists of MCU Unit, Button, LTE Module with GNSS, CAN, Buzzer antenna and Corresponding Power Supply Unit.

The scope includes Schematics Design, PCB Design, mechanical design and Firmware development.

This document will cover key project milestones for Hardware design, mechanical design, Firmware development, Prototyping and Testing. Trinova AI welcomes any custom design request and can provide the necessary expertise, experience, and fast turnaround to meet your special requirements.

The purpose of this Statement of Work (SoW) is to define the below points in details.

1. Client Requirement
2. Trinova AI Compliance
3. Trinova AI design approach
4. Milestones and Schedule
5. Deliverables

Block Diagram:

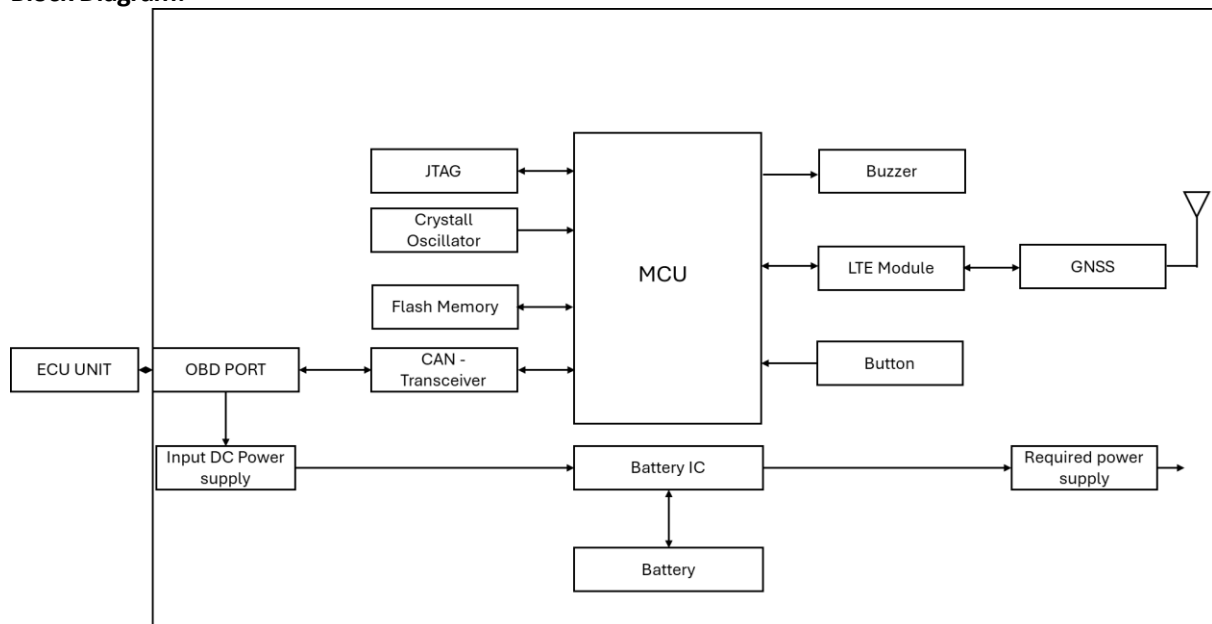


Figure 1: Block Diagram of Project

2.Requirements

Hardware Specification:

Sr. No	Parameter	Description	Compliance
1	Power supply	To power up the device	Yes
2	Battery	To power up the device continuously, even main power not available.	Yes
3	MCU	STM32 To control entire functionality.	Yes
4	JTAG	To program the board	Yes
5	Crystal Oscillator	To generate the clock signals to the MCU.	Yes
6	Buzzer	To specify the theft indication.	Yes
7	Button	To reset the device.	Yes
8	GNSS	To get the location details	Yes
9	LTE module	To send the data to the cloud.	Yes
10	CAN Transceiver	To get the vehicle health details.	Yes

Table 1: Hardware specifications of the Project

Firmware Specification:

Sr. No.	Parameter	Specification from Client	Compliance
1	Firmware	1. System Power-On & Initialization On vehicle power-up, the device initializes: • Microcontroller and system clocks • Power management and voltage monitoring • External flash memory • LTE communication module • GNSS (GPS) subsystem • CAN (OBD-II) interface • Security, tamper, and buzzer interfaces Self-diagnostics and hardware health checks are performed. 2. Communication & Protocol Initialization The firmware configures the LTE module with network parameters and secure cloud channels. The CAN protocol (ISO 15765-4) is initialized for vehicle data acquisition. GNSS subsystem begins satellite acquisition in parallel. 3. Vehicle Data Acquisition The CAN interface retrieves supported OBD-II parameters such as ignition status, speed,	Yes

			engine RPM (if supported), battery voltage, and diagnostic indicators. All data is validated and timestamped. 4. Cloud Connection & Initial Data Upload The LTE module connects to the cloud server and authenticates the device. Device status and GNSS data are transmitted to confirm readiness. 5. Network Failure Handling & Data Buffering During connectivity loss, tracking records and events are stored in external flash memory. Data is uploaded in sequence once connectivity is restored. 6. Trip Detection & Live Tracking Mode Trip start is detected automatically based on ignition or movement. Live tracking data is transmitted at configurable intervals. 7. Parking Mode Operation Parking mode is entered automatically upon ignition OFF. Location updates are sent at user-configured intervals. 8. Theft & Tamper Detection The firmware monitors OBD disconnection, power loss, and unauthorized movement. Upon detection, the buzzer is activated and alerts are sent to the cloud. 9. Remote Control & Safety Logic Remote commands are processed with safety validation. Engine immobilization is executed only when speed is zero and ignition is OFF. 10. OTA Firmware Update Support The device supports secure OTA updates with integrity checks, scheduled deployment, and rollback protection. 11. Continuous Health Monitoring The firmware continuously monitors power, communication health,	
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		memory usage, and system integrity, reporting status to the cloud.	
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Table 2: Firmware specifications of the Project

Mechanical Design:

- As per the specified size.

Open Issues:

S. No.	Open Date	Open Issue	Status	Action	Close Date

Design Approach

- [Design Approach](#)

In this section, design approach for the project is explained. The following are the main blocks in our designing approach.

- Client Requirements
- Documentation (Schematics, BOM, Layout, and Gerber)
- Schematic Drawing
- PCB Designing
- Mechanical design
- Firmware Development
- Testing
- Deliver to Client

- [Bill Of materials:](#)

The following are the main components in Design.

S.No.	Description	Quantity
1	Power supply unit	1
2	MCU	1
3	JTAG	1
4	Crystall Oscillator	1
5	LTE Module	1

6	Flash memory	1
7	Button	1
8	Antenna	1
9	CAN Transceiver	1
10	Buzzer	1
11	Battery	1

Table 3: Bill of Materials

- Tools Required

The following tools are required for design, analysis and testing of the project PC.

S.No.	Tool	No. of Licenses	Purpose
1	KiCad	1	Schematic designing
2	KiCad	1	PCB Layout designing
3	Oscilloscope	1	To observe signals
4	Power supply	1	To feed input

Table 4: List of Tools Required for Project

Note: The tools suggested under subject to availability. During the design, Trinova AI may alter the design & Testing tools. No extra charges for the design tools license fee. Trinova AI have the basic testing equipment includes Power Supply (0-30 VDC), Oscilloscope (0-100 MHz), Function Generator (0-30 MHz) & Multimeter. If testing requires a special equipment, Trinova AI will charge as per actuals.

3. Project Execution Procedure

- Trinova AI's Responsibilities

Trinova AI Technologies Pvt. Ltd. will be responsible for providing following items:

- Specification Analysis
- Specification finalization
- Schematic and PCB Design
- Approval for each milestone
- Technical clarifications
- Technical support if required
- Documentation
- Connectors and their placement can be mutually agreed upon

- Clients Responsibilities

Client will be responsible for providing following items:

- Feedback and Approval on each milestone
- Technical clarification to Trinova AI Team
- Procurement support during Design, Prototyping and Production.
- Connectors placement suggestion
- Documentation approval

- **Reporting**

Project status reporting occurs throughout the duration of the Project. On a weekly basis, the Trinova AI team will send out a status report to Client. These weekly reports will track all action items, open issues, risks or any changes to program plan. The Project report will be a brief document containing:

- Accomplishments for the previous week
- Objectives for the coming week
- Overall schedule status
- Major issues to be resolved/surfaced

This will be followed by a weekly conference call to review the status report as well as issues.

- **Change Management**

Changes to the project scope and requirements can lead to change requests. Trinova AI will perform an impact analysis of the change and impact on project schedules and costs and follow the change-management process referred to in the RFQ for further processing.

Any changes from what is mentioned in the scope of work will need a discussion and follow the change management process. The changes will be implemented only after obtaining approval from Client. If, during the course of the project, a decision is made to deviate from the expected cost, and that incurs additional cost, a mail confirmation is required from both parties to move forward with the change in the requirement with new commercials.

- **Limitation of Liability and Refund Policy**

In the event that the project does not succeed due to insufficient vendor support, Requirement deviation due to a technical concern, delayed responses from the client, or lack of support from the client, Trinova AI shall not be held liable for the project failure or delay, and the client shall not be entitled to any refunds, unless Trinova AI declines to continue the project. In such cases, the refund amount shall be equivalent to the value of the outstanding work and shall be processed within 90 days of Trinova AI rejection to continue the project date. This clause shall survive the termination of this SoW.

If the client is unresponsive or inactive for a period of 45 consecutive days from the date of the last email communication, the SOW shall be deemed terminated automatically. In such cases, no refunds shall be initiated by Trinova AI.

- **Outsourcing or Third-party sourcing**

Product Design, Prototyping and Production involves many steps. Some of these steps may fall outside of Trinova AI capabilities. This document serves as an authorization for Trinova AI to outsource certain steps in the product design, prototyping and Production process to a third party.

The client acknowledges that the outsourcing of certain steps in the product design and prototyping process may require the disclosure of confidential information to a third party. The client agrees that Trinova AI shall be permitted to disclose such confidential information to the third party for the sole purpose of completing the product design and prototyping process. Trinova AI shall take all reasonable

measures to ensure the highest level of data security in connection with the product design, prototyping and Production process during outsourcing.

Project Schedule (Project kick-off + Numbers in the chart refer to weeks)

MILESTONE	Start	Completion
MS1: Project Approval	Project Kick-off	Once schedule approved by Client, the project will be started.
MS2: Analysis & Finalization of details of hardware, Firmware and Mechanical design	Project Kick-off+1	Analysis of the hardware and selection of the components
MS3: Preparation of schematic	Project Kick – off+2	Circuit design and schematic entry in the KiCad Tool, net list verification and BOM generation and reviews
MS4: PCB Designing	Project Kick – off +4	Layer Stack up finalization Placement of the components on board, layout design, Gerber generation and reviews
MS5: Component Sourcing	Project Kick – off +6	Purchase of components
MS6: Fabrication	Project Kick – off +6	Fabrication of the PCB
MS7: Assembly	Project Kick – off +7	Assembly of physical components on the board
MS8: Testing of PCBs	Project Kick – off +8	Complete device brings up with power supply, communication with MCU through JTAG Emulator and with the firmware.
MS9: Firmware and Mechanical design	Project Kick – off+9	Firmware development and mechanical design.
MS10: Preparation of ATP	Project Kick – off +12	Preparation of documents for ATP
MS11: Documentation	Project Kick – off +13	Documentation for the design and development
MS12: Performance evaluation and Acceptance	Project Kick – off +14	Performance test and acceptance from the Client

4.Deliverables

- Deliverables from Trinova AI

s.no	Deliverables	Description
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1	The project	Design Files
2	Acceptance Test Procedure Document	1 Set

Table 5: Deliverables from Trinova AI

Note:

- The Quotation is only for design of Hardware and Firmware development. After design completion Trinova AI will share separate quotation for Manufacturing the PCBs as per the client request.
- If the project requires a bench setup, the client shall be responsible for covering the cost of any necessary development kits. In the event that the client rejects the suggested bench setup, the project timelines may be extended accordingly, and there is a risk of failure since the attempt will be direct. The suggested bench setup will be determined based on the availability of development kits.
- In the event that development kits are either not available or have a significant lead time, there may be a possibility of avoiding bench setups.
- Prerequisite to start the work: PO/Work Order along with advance payment
- All payments to be made in the name of Trinova AI Technologies Pvt Ltd
- Service Tax charged as applicable if paid in Indian currency (18% GST), there is no service tax for foreign remittance
- Travel cost, boarding, lodging (if outside Bangalore) on account of Design support, and second batch production will be borne by the client and will be billed on actuals.
- If any requirement changes from the client during the development stage, Trinova AI will share a new effort estimation along with new timelines.
- The lead time for the development or manufacturing orders may be varied depending on the communication between Trinova AI and clients during the course of development or manufacturing. The response time from the clients or their vendors will not add up in the lead time. Lead time starts from the date of clarified data, valid PO & advance, whichever is later.
- 'COVID-19' delay needs to be accommodated.
- Gerber Generation and Validation require a lot of effort, if any changes require kindly let us know during the time of approval itself. Once the Gerber is generated and if any changes are required, Trinova AI will share the new effort estimation accordingly.
- Once the Gerber is shared with the client, Trinova AI will share an estimation for BoM Procurement, PCB Fabrication, PCB Assembly, and Stencil.
- Once Trinova AI receives an official confirmation along with advance payment from the client, Trinova AI will start Procurement of Components and Manufacturing the boards.
- Semiconductor stocks are volatile there may be an unexpected delay may occur, during components procurement. Trinova AI Team will inform to Client promptly about the delivery schedule.
- Once Trinova AI starts Manufacturing, the manufacturing process can't be stopped. Design or Manufacturing orders are Non-Cancellable and Non-Returnable (NCNR).

- In case if required to go for second or other revisions with in the scope, Trinova AI will support free of cost for the design changes and Trinova AI will submit new quotation for manufacturing of further revisions.
- Packing and forwarding charges extra
- All Disputes are subject to Bangalore Jurisdiction only
- **Acceptance Criteria**
- For all deliverables, Client will have 5 business days (1 week) to accept the deliverable or in case of rejection of the deliverable.
- **Production support**
- Trinova AI can support mass Production.
- Trinova AI can give end to end product design and manufacturing support such that Trinova AI is maintaining quality of product. Trinova AI Team will share separate quotation for production after design completion.

5.Management & Technical Requirements

- **Project Management**

Trinova AI will provide a Project Manager for management and direction of the Trinova AI activities to resolve problems that will arrive during execution of SOW. The Project Manager is responsible for maintaining an adequate system for tracking the progress of this effort and bring it to completion and maintaining workable interfaces and relationships with concurrent and any subsequent associated contracts.

- **Technical Management**

Trinova AI shall provide a Project Manager responsible for the technical management of the contract requirements and coordination of these efforts with Client.

6.Resourcing and Responsibilities

Name	Organization	Designation	Mail ID
Sushmitha	Trinova AI Technologies Pvt Ltd	Director	sales@trinovaaitech.com
Sudha Mallesh	Trinova AI Technologies Pvt Ltd	Director	technical@trinovaaitech.com